Data Source Report

Task 1: Data Collection & Data Source Identification

Submitted By	Arshpreet Singh
Objective	Identify, collect, and validate reliable datasets for data analytics.
Project Focus	Climate change, carbon emissions, energy transition, and environmental trends.

Introduction

Data collection is an important step in the data analytics process. Reliable and well-structured data helps analysts generate accurate insights and make informed decisions. For this task, multiple climate and energy-related datasets were collected and examined.

Dataset Information

No.	Dataset	File Name	Format	Records	Columns	Description
1	CO2 Emissions Dataset	co2_emissions_yearly.csv	CSV	1350	8	Yearly carbon dioxide emission statistics for different countries and regions.
2	Energy Mix Dataset	energy_mix_yearly.csv	CSV	1350	14	Energy production from coal, oil, gas, nuclear, hydro, solar, wind, and renewable sources.
3	Temperature Anomaly	temperature_anomaly_monthly.csv	CSV	2528	7	Monthly temperature

	Dataset					anomaly measurements and climate indicators.
4	Carbon Prices Dataset	carbon_prices_daily.csv	CSV	15866	5	Daily carbon market prices used for environmental policy and carbon trading analysis.
5	Climate Events Dataset	climate_events.csv	CSV	50	11	Major climate-related events and disasters with severity information.

Data Source Type

The datasets used in this project are examples of secondary data because they were collected and prepared by external sources before being used for analysis.

Primary Data

Primary data is collected directly by researchers through surveys, interviews, observations, or experiments.

Secondary Data

Secondary data is data that has already been collected and published by organizations, researchers, or public repositories.

Data Formats

The datasets are stored in CSV (Comma-Separated Values) format. CSV files are widely used because they are easy to read, share, and analyze using spreadsheet and data analytics tools.

Data Validation

- Structured tabular format
- Consistent column names
- Suitable for analysis and visualization
- Large number of records
- Relevant to climate and energy analytics
- Easy to process using analytical tools

Why These Datasets Are Reliable

- Well-structured and organized data
- Historical records covering multiple years
- Suitable for educational and analytical purposes
- Contains measurable and quantifiable information
- Can be used for trend analysis, visualization, and reporting

Conclusion

The selected climate and energy datasets provide valuable information for studying environmental trends, carbon emissions, renewable energy adoption, carbon pricing, and climate-related events. These datasets are reliable, structured, and suitable for data analytics projects. Through this task, I learned how to identify data sources, validate datasets, and organize data for future analysis.

Learning Outcomes

- Understanding Primary and Secondary Data
- Data Source Identification
- Data Collection Techniques
- Data Validation Methods
- Working with CSV Datasets
- GitHub Project Documentation